

Computer System Architecture

Tutorial 2:

1. A benchmark program is run on an 80 MHz processor. The executed program consists of 100,000 instruction execution, with the following instruction mix and clock cycle count:

Instruction Type	Instruction Count	Cycle per Instruction
Integer Arithmetic	45000	1
Data Transfer	32000	2
Floating Point	15000	2
Control Transfer	8000	2

Determine the effective CPI, MIPS rate, and execution time for this program

2. The performance of a 100MHz microprocessor P is measured by executing 10,000,000 instruction of benchmark code, which is found to take 0.25s. What are the values of CPI and MIPS for this performance experiment? Is P likely to be superscalar?
3. Suppose that a single-chip microprocessor P operating at clock frequency of 50MHz is replaced by a new model P', which has **the same architecture** as P but has a clock frequency of 75MHz.
 - a. If P has a performance rating of p MIPS for a particular benchmark program Q, what is the corresponding MIPS rating p' for P
 - b. P takes 250s to execute Q in a particular personal computer system C. On replacing P by P' in C, the execution time of Q drops only to 220s. Suggest a possible reason for this disappointing performance improvement.
4. Suppose we have two implementations of the same instruction set architecture. Computer A has a clock cycle time of 250 ps and a CPI of 2.0 for a given benchmark program, and computer B has a clock cycle time of 500 ps and a CPI of 1.2 for the same program. Which computer is faster for this program and by how much?

5. A compiler designer is deciding between two codes for a particular machine. Based on the hardware implementation, there are three different classes of instructions: Class A, Class B, and Class C, and they require one, two, and three cycles respectively.

First code has 5 instructions: 2 of A, 1 of B, and 2 of C.

Second code has 6 instructions: 4 of A, 1 of B, and 1 of C.

Which code is faster?

By how much?

What is the CPI for each code?

$$\begin{aligned}
 \text{CPU time} &= \text{CPU clock cycle} \times \text{clock cycle time} \\
 &= \frac{\text{CPU clock cycle}}{\text{clock rate}} \\
 &= \text{Instruction count} \times \text{CPI} \times \text{clock cycle time} \\
 &= \frac{\text{Instruction count} \times \text{CPI}}{\text{clock rate}}
 \end{aligned}$$

$$\begin{aligned}
 \text{CPU clock cycle} &= (\text{Instruction / program}) \times (\text{clock cycles / instruction}) \\
 &= \text{Instruction count} \times \text{CPI}
 \end{aligned}$$

$$\frac{\text{CPU time old}}{\text{CPU time new}} = \frac{\text{clock rate new}}{\text{clock rate old}}$$

$$\text{CPI} = \sum_{i=1}^n \text{CPI}_i \times F_i$$

$$\text{MIPS} = \frac{\text{Instruction count}}{\text{execution time} \times 10^6}$$

$$\text{MFLOPS} = \frac{\text{Floating point operation}}{\text{execution time} \times 10^6}$$

Tutorial 2:

1. A benchmark program is run on an 80 MHz processor. The executed program consists of $100,000$ instruction execution, with the following instruction mix and clock cycle count:

Instruction Type	Instruction Count	f	Cycle per Instruction CPI
Integer Arithmetic	45000	0.45	1
Data Transfer	32000	0.32	2
Floating Point	15000	0.15	2
Control Transfer	8000	0.08	2

Determine the effective CPI, MIPS rate, and execution time for this program

$$\text{effective CPI} = (0.45 \times 1) + (0.32 \times 2) + (0.15 \times 2) + (0.08 \times 2) = 1.55$$

$$\begin{aligned} \text{execution time (CPU time)} &= \text{CPU clock cycle} \times \text{CPU clock time} \\ &= \text{Instruction count} \times \text{CPI} \times \frac{1}{f} \\ &= 100\,000 \times 1.55 \times \frac{1}{80 \times 10^6} \\ &= 1.9375 \times 10^{-3} \text{ s} \\ &= 1.9375 \text{ ms} \end{aligned}$$

$$\text{MIPS rate} = \frac{100\,000}{1.9375 \times 10^{-3} \times 10^6} = 51.613$$

$$\begin{aligned} \text{a) effective CPI} &= \frac{\text{total no cycle}}{\text{total no instruction}} \\ &= \frac{155\text{k}}{100\text{k}} \\ &= 1.55 \end{aligned}$$

$$\begin{aligned} \text{b) execution time} &= \text{total cycle} \times t \\ &= \frac{\text{total cycle}}{f} \\ &= \frac{155\text{k}}{80\text{MHz}} \\ &= 1.94 \text{ ms} \end{aligned}$$

$$\begin{aligned} \text{c) MIPS} &= \frac{\text{no of instruction}}{t \text{ (in M)}} \\ &= \frac{100\text{k}}{(1.94 \text{ ms} \times 10^{-3})} \\ &= 51.55 \end{aligned}$$

$$\begin{aligned} \text{CPU time} &= \text{CPU clock cycle} \times \text{clock cycle time} \\ &= \frac{\text{CPU clock cycle}}{\text{clock rate}} \\ &= \text{Instruction count} \times \text{CPI} \times \text{clock cycle time} \\ &= \frac{\text{Instruction count} \times \text{CPI}}{\text{clock rate}} \end{aligned}$$

$$\begin{aligned} \text{CPU clock cycle} &= (\text{Instruction / program}) \times (\text{clock cycles / instruction}) \\ &= \text{Instruction count} \times \text{CPI} \end{aligned}$$

$$\begin{aligned} \text{CPU time old} &= \frac{\text{clock rate new}}{\text{clock rate old}} \\ \text{CPU time new} &= \frac{\text{clock rate old}}{\text{clock rate new}} \end{aligned}$$

$$CPI = \frac{f}{f_{\text{old}}} \times CPI_{\text{old}}$$

$$\text{MIPS} = \frac{\text{Instruction count}}{\text{Execution time} \times 10^6}$$

$$\text{MFLOPS} = \frac{\text{Floating Point Operation}}{\text{Execution time} \times 10^6}$$

$$45\text{k} \times 1 = 45$$

$$32\text{k} \times 2 = 64$$

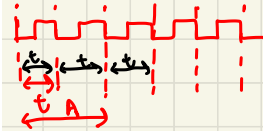
$$15\text{k} \times 2 = 30$$

$$8\text{k} \times 2 = 16$$

$$155\text{k}$$

$$f = 80\text{MHz}$$

$$100\text{k instructions}$$



If instruction require 2 CPI (2 clock) . A

3 instruction, 3 CPI, number of execution time = cycle $\times t$

$$T = \frac{1}{f}$$

$$f = \frac{1}{T}$$

2. The performance of a 100MHz microprocessor P is measured by executing $10,000,000$ instruction of benchmark code, which is found to take 0.25s . What are the values of CPI and MIPS for this performance experiment? Is P likely to be superscalar?

I_c = Instruction count

f = CPU rate

t = execution time / CPU time

$$I_c = 10,000,000$$

$$f = 100\text{MHz}$$

$$t = 0.25\text{s}$$

$$t = I_c \times \text{CPI} / f = \frac{\text{Instruction count} \times \text{CPI}}{\text{clock rate}}$$

$$0.25 = 10,000,000 \times \text{CPI} / 100 \times 10^6$$

$$\text{CPI} = 2.5$$

$$\text{MIPS} = \frac{\text{Instruction count}}{\text{execution time} \times 10^6}$$

$$= \frac{10,000,000}{0.25 \times 10^6}$$

$$= 40$$

$$f = 100\text{MHz}$$

$$10 \text{ million instruction}$$

$$t = 0.25\text{s}$$

$$\text{CPI} = \frac{\text{tot no of cycle}}{\text{tot no of instruction}}$$

$\therefore$ No, P is not a superscalar because the CPI is greater than 1. A superscalar processor can perform more than one instruction in every cycle and has CPI less than 1.

CPU time (in seconds)

= execution time

= CPU clock cycles $\times$ clock cycle time

= CPU clock cycles / clock rate

= Instruction count $\times$ CPI / clock rate

$$\therefore \text{CPI} = \frac{\text{CPU time} \times \text{clock rate}}{\text{Instruction count}}$$

$$= \frac{0.25\text{s} \times 100\text{MHz}}{10\text{M}}$$

$$= 2.5 \text{ cycles / instruction}$$

Million instruction per s

$$\text{MIPS} = \frac{\text{Instruction count}}{\text{execution time} \times 10^6}$$

$$= \frac{10\text{M}}{0.25 \times 10^6}$$

$$= 40$$

P is likely superscalar as $\text{MIPS} \geq 1$

3. Suppose that a single-chip microprocessor P operating at clock frequency of 50MHz is replaced by a new model P', which has **the same architecture as P but has a clock frequency of 75MHz.**

- If P has a performance rating of p MIPS for a particular benchmark program Q, what is the corresponding MIPS rating p' for P
- P takes 250s to execute Q in a particular personal computer system C. On replacing P by P' in C, the execution time of Q drops only to 220s. Suggest a possible reason for this disappointing performance improvement.

$$\begin{aligned}
 \text{a) } \text{MIPS} &= \frac{\text{Instruction count}}{\text{execution time} \times 10^6} \\
 &= \frac{\text{Instruction count}}{\text{instruction count} \times \text{CPI} / \text{clock rate} \times 10^6} \\
 &= \frac{\text{clock rate}}{\text{CPI} \times 10^6}
 \end{aligned}$$

$$\text{MIPS} = \frac{\text{Instruction count}}{\text{execution time} \times 10^6}$$

$$= \text{Instruction count} / (\text{Instruction count} \times \text{CPI}) / \text{clock rate} \times 10^6$$

$$p = \frac{50}{\text{CPI} \times 10^6}$$

$$p' = \frac{75}{\text{CPI} \times 10^6}$$

$$\text{CPI} = \frac{50}{p \times 10^6}$$

$$= \frac{75}{\frac{50}{p \times 10^6} \times 10^6}$$

$$p' = p \times 1.5$$

$$\begin{aligned}
 \frac{\text{MIPS } p'}{\text{MIPS } p} &= \frac{\text{clock rate } p'}{\text{clock rate } p} \\
 \text{MIPS } p' &= \frac{\text{clock rate } p'}{\text{clock rate } p} \times \text{MIPS } p \\
 \text{MIPS } p' &= \frac{75 \times 10^6}{50 \times 10^6} \times p \\
 &= 1.5p
 \end{aligned}$$

$\therefore$ MIPS of the new model p' is 1.5 times the original model p

- The execution time is limited by other architecture factors such as memory
- The performance improvement is limited by other system bottleneck, such as memory bandwidth, I/O bandwidth

Some possible reasons include:

1. **Memory bottleneck:** The performance improvement may be limited by the memory subsystem's ability to keep up with the faster processor. If the memory cannot provide data and instructions at a rate that matches the increased speed of the processor, the overall performance improvement may be limited.
2. **Instruction-level parallelism limitations:** Even though the clock frequency is increased, if the benchmark program does not have enough independent instructions to execute simultaneously (low instruction-level parallelism), the performance improvement may be limited.

3. **I/O or external dependencies:** The benchmark program may have dependencies on external factors such as I/O operations, disk access, or network latency, which are not affected by the processor's speed increase.

4. **Software optimization:** The benchmark program may not be fully optimized to take advantage of the increased processor speed. Inefficient algorithms or code structures may limit the overall performance improvement.

5. **System architecture limitations:** Other components of the system, such as buses, caches, or interconnects, may become bottlenecks that limit the overall performance improvement.

Overall, it's essential to consider various factors beyond just the processor's clock frequency when evaluating performance improvements.

4. Suppose we have two implementations of the same instruction set architecture. Computer A has a clock cycle time of 250 ps and a CPI of 2.0 for a given benchmark program, and computer B has a clock cycle time of 500 ps and a CPI of 1.2 for the same program. Which computer is faster for this program and by how much?

$$T_A = \text{instruction count} \times 2.0 \times 250 \times 10^{-12}$$

$$T_B = \text{instruction count} \times 1.2 \times 500 \times 10^{-12}$$

$$\frac{T_A}{T_B} = \frac{2.0 \times 250 \times 10^{-12}}{1.2 \times 500 \times 10^{-12}}$$

$$= \frac{5}{6}$$

$$T_A = \frac{5}{6} T_B$$

$$= 1.2 T_B$$

computer A is 1.2 times faster than computer B

(A)

250ps
CPI=2

for one instruction,
2 cycle

execution time = 2 cycle $\times$ 250
= 500ps (better)

(B)

(500ps per cycle
CPI=1.2

1.2 cycle $\times$ 500
= 600ps

$$\frac{B}{A} = \frac{600ps}{500ps}$$

$$= 1.2 \text{ faster}$$

(C)

clock cycle time A = 250ps CPI_A = 2
clock cycle time B = 500ps CPI_B = 1.2
Assume the instruction count = Q
CPU time (in seconds) = execution time

= CPU clock cycle $\times$ clock cycle time
= instruction count $\times$ CPI $\times$ clock cycle time

$$\text{CPU time}_A = Q \times 2 \times 250ps$$

$$= Q \times (5 \times 10^{-11})$$

$$\text{CPU time}_B = Q \times 1.2 \times 500ps$$

$$= Q \times (6 \times 10^{-11})$$

$$\frac{B}{A} = \frac{600ps}{500ps}$$

$$= 1.2 \text{ faster}$$

$$\text{CPU time} = \text{CPU clock cycle} \times \text{clock cycle time}$$

$$= \text{CPU clock cycle} \times \text{clock rate}$$

$$= \text{instruction count} \times \text{CPI} \times \text{clock cycle time}$$

$$= \text{instruction count} \times \text{CPI} \times \frac{1}{\text{clock rate}}$$

$$\text{CPU clock cycle} = (\text{instruction/program}) \times (\text{clock cycles/instruction})$$

$$= \text{instruction count} \times \text{CPI}$$

$$\frac{\text{CPU time old}}{\text{CPU time new}} = \frac{\text{clock rate new}}{\text{clock rate old}}$$

$$\text{CPI} = \frac{2}{1} \times \text{CPI}_1 \times \text{FI}$$

$$\text{MIPS} = \frac{\text{instruction count}}{\text{execution time} \times 10^6}$$

$$\text{MFLOPS} = \frac{\text{floating point operation}}{\text{execution time} \times 10^6}$$

5. A compiler designer is deciding between two codes for a particular machine. Based on the hardware implementation, there are three different classes of instructions: Class A, Class B, and Class C, and they require one, two, and three cycles respectively.

First code has 5 instructions: 2 of A, 1 of B, and 2 of C.

Second code has 6 instructions: 4 of A, 1 of B, and 1 of C.

Which code is faster?

By how much?

What is the CPI for each code?

Instruction	F	CPI
A	$\frac{2}{5}$	1
B	$\frac{1}{5}$	2
C	$\frac{2}{5}$	3

$$\begin{aligned} \text{CPI} &= \left(\frac{2}{5}\right) + \left(2 \times \frac{1}{5}\right) + \left(\frac{2}{5} \times 3\right) \\ &= 2 \end{aligned}$$

Instruction	F	CPI
A	$\frac{4}{6}$	1
B	$\frac{1}{6}$	2
C	$\frac{1}{6}$	3

$$\begin{aligned} \text{CPI} &= \left(\frac{4}{6}\right) + \left(\frac{2}{6}\right) + \frac{3}{6} \\ &= 1.5 \end{aligned}$$

$T = \text{instruction count} \times \text{CPI} \times \text{clock cycle time}$

$T_1 = 5 \times 2 \times \text{clock cycle time}$

$T_2 = 6 \times 1.5 \times \text{clock cycle time}$

$$\frac{T_1}{T_2} = \frac{10}{9}$$

$$T_2 = \frac{9}{10} T_1$$

$\therefore$ The second code is faster than the first code by 1.1 times.

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First code has 5 instructions: 2 of A, 1 of B, and 2 of C.

Second code has 6 instructions: 4 of A, 1 of B, and 1 of C.

Which code is faster?

By how much?

What is the CPI for each code?

①

A	2	x 1	= 2
B	1	x 2	= 2
C	2	x 3	= 6

require 10 cycle

cycle per instruction

$$CPI = \frac{10}{5}$$

$$= 2$$

②

A	4	x 1	= 4
B	1	x 2	= 2
C	1	x 3	= 3

9 cycle

faster

$$\frac{10}{9} = 1.11 \text{ faster}$$

$$CPI = \frac{9}{6}$$

$$= 1.5$$

$$CPI \text{ for 1st} = \frac{2 + (1 \times 2) + (2 \times 3)}{5}$$

$$= 2$$

$$\text{Total cycle A} = 2 \times 5 = 10$$

$$\text{Total cycle B} = 1.5 \times 6 = 9$$

A is slightly faster by 11%

$$\text{Execution time A} = \text{total cycle} \times t = 10t$$

$$B = \text{total cycle} \times t = 9t$$

$$B \text{ is faster by } \frac{10t}{9t} = 1.11$$